

SM SALA 5.5 20, 21, 24, 25

(20) REGRA DA POTENCIA

$$\int_1^{100} x^{100} dx = \frac{x^{100+1}}{100+1} = \frac{1}{101} \left| \begin{array}{c|c} x^{101} & 1 \\ \hline 101 & 10 \end{array} \right|$$

(21) $\int_1^4 (5 - 2t + 3t^2) dt$

REGRA DA CONSTANTE $\int k dt = kt$

$$F(t) = (5t - t^2 + t^3) \Big|_1^4 \Rightarrow F(4) = 5(4) - (4)^2 + (4)^3$$

$F(4) = 68$ $F(1) = 5(1) - (1)^2 + (1)^3$
 $F = 5$

B-A

$$68 - 5 = 63$$

(24) $\int_1^8 \sqrt[3]{x} dx \Rightarrow \int_1^8 x^{1/3} dx = \frac{x^{1/3+3/3}}{1/3+3/3} = \frac{3}{4} x^{4/3} \Big|_1^8$

$$\frac{3}{4} \cdot (8^{4/3} - 1^{4/3}) = \frac{3}{4} \cdot (16 - 1) = \frac{3}{4} \cdot (15)$$

$8^{4/3} = 16$
 $1^{4/3} = 1$

(25) $\int_1^2 \frac{3}{t^4} dt \Rightarrow \int_1^2 3t^{-4} dt = \frac{t^{-4+1}}{-4+1} = \frac{3}{-3} \cdot \frac{1}{t^3} = -\frac{1}{t^3}$

6x6 - 4 + 1 = -3

$$F(2) = \frac{1}{2^3} = \frac{1}{8}$$

$$3 \cdot -1 = -3$$

$$F(1) = \frac{1}{1^3} = 1$$

F(A-B)

$$\left(-\frac{1}{8} \right) - (1) = -\frac{1}{8} - 1 = -\frac{1}{8} - \frac{8}{8} = -\frac{9}{8}$$

1/1

Ex SA A 5.5 26, 29, 35

(26) $\int_0^{2\pi} \cos x = [\sin x]_0^{2\pi}$

$\cos x dx = \sin x$

$f(2\pi) = \sin(2\pi) = 0$

$0 - 0 = 0$

$f(0) = \sin(0) = 0$

$A - B = 0$

(21) $\int_1^9 \frac{x-1}{\sqrt{x}} dx = \int_1^9 \frac{x}{\sqrt{x}} - \frac{1}{\sqrt{x}} = \int_1^9 x^{1/2} - x^{-1/2}$

$\sqrt{x} = x^{1/2}$

$x^{1/2} = x^{-1/2} = x^{-1/2} = \frac{x^{n+1}}{n+1} = \frac{x^{3/2}}{3/2}$

$= \frac{2}{3} x^{3/2} - 2 x^{1/2}$

$x^{-1/2} = -x^{1/2} = -2x^{1/2}$

Primitiva $F(x) = \frac{2}{3} x^{3/2} - 2x^{1/2}$

$F(9) = \frac{2}{3} (9)^{3/2} - 2(9)^{1/2} = \frac{2}{3} (27) - 2(3) = 18 - 6 = 12$

$\sqrt{9} = 3$

$9^{3/2} = (3)^3 = 27$

$F(1) = \frac{2}{3} (1)^{3/2} - 2(1)^{1/2} = \frac{2}{3} (1) - 2(1) = \frac{2}{3} - 2 = -\frac{4}{3}$

$A - B$

$12 - (-\frac{4}{3}) = 12 + \frac{4}{3} = \frac{36}{3} + \frac{4}{3} = \frac{40}{3}$

(35) $\int_1^2 \frac{v^3 + 3v^2 + 2v}{v^4} dv = \int_1^2 \frac{1}{v} + 3\frac{v^2}{v^4} + 2\frac{v}{v^4}$

$\frac{1}{x} = \ln(x)$

$f(v) = \ln(v) + v^{-3} + v^{-3}$

$A - B$
 $\ln(2) + \frac{1}{8} - (\ln(1) + \frac{1}{8})$

$3v^2 = 3 \cdot \frac{v^3}{3}$

$f(2) = \ln(2) + 2^{-3} = \ln(2) + \frac{1}{8}$

$f(1) = \ln(1) + 1^{-3} = 0 + 1 = 1$

tilibra